

Investigating the lurking mechanism of SNS users: a comprehensive examination of context-specific cues and role stresses

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Abstract

Purpose – Social networking services (SNS) empower users with a robust capability to connect with others and manage their social relationships. However, as the size of users' social networks increases, coupled with the inherent boundary-spanning technical features of SNS, users are faced with unprecedented role stresses. This, in turn, leads to maladaptive lurking decisions. This study delves into the mechanism of this technology-induced decision-making process among SNS users.

Design/methodology/approach – Survey data were collected from 491 Chinese WeChat Moment users. The model and hypotheses testing were conducted using SmartPLS 4.0.

Findings – Our findings indicate that both social network size and boundary spanning have a positive influence on role conflict and role overload. Both role conflict and role overload significantly contribute to SNS fatigue, which further intensifies users' lurking intention. Furthermore, SNS fatigue fully mediated the relationship between role conflict and lurking intention, and partially mediated the relationship between role overload and lurking intention.

Originality/value – Our study offers a fresh viewpoint for comprehending lurking behaviors on SNS, furnishing practical insights for platform providers. Additionally, it paves the way for future research into the deeper mechanisms driving SNS lurking behaviors, by providing a novel construct (i.e. boundary spanning) to distinguish and measure the unique social environment of SNS.

Keywords SNS, Lurking intention, Boundary spanning, Role stresses, SNS fatigue

Paper type Research paper

1. Introduction

Social networking services (SNS) have become an integral part of our daily lives, transforming the way we communicate and interact with each other. Prominent SNS such as Facebook and WeChat Moment (WM) allow users to share insights into their lives, express their opinions and update their status within their social networks. According to a recent report [Tencent \(2022\)](#), WM, the most widely used SNS in China, possesses approximately 1.31 billion monthly active users. Meanwhile, Facebook, with its global reach, has the largest user base in the world, with around 2.96 billion monthly active users ([Meta, 2022](#)). The sheer



size of these user bases, which continues to grow, has transformed SNS into not just a tool for managing social connections but also a primary channel for disseminating various types of information. The Digital 2021 Global Overview Report (KEMP, 2021) further substantiated this by revealing that people spend an average of 6 h and 54 min per day using the Internet across all devices, with 2 h and 25 min of that time dedicated to using SNS. This significant attention from users has led to substantial economic value and has had a profound impact on people's lives.

The pervasive use of SNS in daily life has significantly expanded and complicated people's social networks. Many offline social connections are transitioning online, with new ones being continually established. As of 2015, the average number of friends per WeChat user is approximately 128, a figure that is expected to grow as more users join the workforce (Tencent, 2015). WeChat has recently expanded the maximum number of contacts a user can have from 5,000 to 10,000. This enhancement reflects the platform's commitment to adapting its network capabilities to meet the growing demands and needs of its users. As users expand their social networks on SNS, their social circles become increasingly varied and diverse. Notably, WeChat has become a hub for most people's social circles, encompassing relatives, friends, supervisors, colleagues, customers and many others. When engaging in social activities across these diverse circles, users must adapt to different roles and behave according to the expectations associated with each role (Zhang *et al.*, 2011). Consequently, a single user may juggle multiple, and often conflicting, roles within their various social circles on an SNS (Zhang *et al.*, 2011). For instance, a user might be a colleague or a boss within a work-related network, while simultaneously being a parent or a child within their family network. In traditional offline social environments, temporal and spatial constraints limit users to playing a single role that fits their specific situation, such as only being a colleague while at the workplace. However, these constraints diminished on SNS (Rui and Stefanone, 2013), leading to a blurring of boundaries between various social roles. As a result, users often must navigate multiple roles simultaneously on SNS. This phenomenon, known as boundary spanning, is common among SNS users and can lead to increased role stress, including role conflict and role overload (Zhang *et al.*, 2011; Liu *et al.*, 2020).

Although current SNS, such as WM and Facebook, enables users to categorize their contacts into distinct groups to mitigate potential role stresses, the effectiveness of this mechanism is undermined by its boundary-spanning nature. The requirement for users to select continuously and manually who can view their posts can be an exhaustive process, particularly as the size and complexity of their contact lists increase. This task demands considerable effort and can lead to user fatigue. In practice, many users do not rigorously organize all their contacts into specific groups. Consequently, lurking has become a less time-consuming and labor-intensive coping strategy for managing these circumstances. Lurking, a type of inactive online user behavior characterized by non-posting (Edelmann, 2013), is a common phenomenon on SNS (Hildebrand, 2017). This is evidenced by the fact that more than 80% of the contents on SNS are created by <20% of the active users, a distribution that aligns with Pareto's 80/20 law (Matei and Bruno, 2015). Users' active participation is the key to maintaining the prosperity of SNS, which generates new knowledge and values for businesses (Jansen *et al.*, 2009), the economy (Asur and Huberman, 2010) and public politics (Tumasjan *et al.*, 2010). For SNS providers, users' active participation significantly contribute to their sustainability. For SNS users, lurking, as a maladaptive coping strategy, can reduce their short-term stress at the expense of increased long-term stress (Wisniewski *et al.*, 2014). As such, it is crucial to understand the underlying mechanisms of how to increase their user engagement, especially for lurkers, and to maintain the well-being of users. Regrettably, few information systems (IS) studies have delved into the mechanisms of lurking behavior within the context-specific cues of SNS, particularly the combined impacts of users' social network size and the boundary-spanning features of SNS.

Existing research on lurking has often justified the phenomenon as reasonable, attributing it to personal traits such as being a loafer or freeloader (Curien *et al.*, 2006), users' low self-efficiency (i.e. being a newbie) (Bogdan and Bowers, 2007; Preece and Shneiderman, 2009), or the fact that users' information needs can be satisfied without posting (Preece *et al.*, 2008; Nonnecke and Preece, 2001; Nonnecke *et al.*, 2004). However, lurking can often be a maladaptive coping strategy for SNS users, especially when their extensive social network size intersects with the boundary-spanning technical feature (Liu *et al.*, 2020). In such situations, users may alleviate their short-term role stresses at the cost of increased long-term psychological discomfort. Contrary to the behaviors associated with contributing content, lurking behavior may negatively affect users' self-expression (Shao, 2009), relationship maintenance (Lee *et al.*, 2014), impression management (Krämer and Winter, 2008) and could even result in financial repercussions for SNS (Park *et al.*, 2016).

To bridge the existing research gap, this study delves into the technology-driven decision-making process of users within the context of SNS. It aims to answer the following research questions: (1) What are the specific contextual cues that lead to maladaptive lurking intention by users within the SNS environment? (2) How do these specific contextual cues trigger perceived role stresses for users on SNS? (3) How do role stresses influence the lurking intention of SNS users? To address these questions, this study anchored in the burnout theoretical model to investigate the impact of SNS context-specific cues on users' lurking intention from a role perspective. Specifically, we considered the specific cues within the SNS context leveraging the social and technical aspects. The context-specific cues of SNS will induce users' cognitive pressures (i.e. role conflict and role overload) and emotional fatigue, which ultimately lead to their lurking intention.

This study offers three significant contributions to the burgeoning literature on SNS lurking. Firstly, we examined the context-specific cues from both social and technical aspects. By integrating these two aspects, we can delve into the impacts of role stresses engendered by the distinctive social environment on SNS, thus expanding our comprehension of platform-based features within the social context of SNS. Secondly, we identified boundary spanning as a second-order construct, comprising two dimensions: parallelism and reprocessability. This provides a novel construct to distinguish and measure the boundary-spanning activities on SNS. Finally, anchored in the burnout theoretical model and integrated with a role perspective, this study stands as one of the pioneering efforts in elucidating the mechanism behind the maladaptive lurking intention observed among SNS users. This contribution further enriches the SNS lurking literature, aiding in a deeper comprehension of the decision-making process behind users' lurking intentions. The integration of role theory and the burnout theory also theoretically extends these two theories.

The structure of the remainder of this paper is as follows. Section 2 introduces the relevant theoretical background. This is followed by the presentation of the proposed research model and hypotheses in Section 3. The research methodology and data analysis are detailed in Sections 4 and 5, respectively. Subsequently, the research findings, contributions, implications and limitations are discussed in Section 6. Finally, Section 7 gives a conclusion of this study.

2. Literature review and theoretical background

2.1 SNS lurking

Lurking is a term associated with non-posting behavior (Edelmann, 2013), defined as a form of inactive online user behavior characterized by infrequent posting (Rafaeli *et al.*, 2004), being silent (Leshed, 2005) or a lack of contribution (Phang *et al.*, 2015). This phenomenon is prevalent in both online communities and SNS, as user behavior online often adheres to

Pareto's law (Matei and Bruno, 2015). Extensive research has been conducted on this phenomenon within the context of online communities, identifying four primary reasons for lurking: environmental influence, personal preference, the relationship between the individual and the group and security considerations (Sun *et al.*, 2014). Specifically, Preece *et al.* (2008) suggested the top five reasons for lurking: the lack of a need to post, the desire to learn more about the group before participating, the belief that their silence is helpful, difficulties with the software (i.e. poor usability), and a dislike of the group dynamics or a sense that the community is not a good fit for them.

Current research on lurking within SNS has limitations, particularly in distinguishing the fundamental differences between SNS and online communities (see Appendix 1). These studies often overlook the impact of changes in the social environment, caused by technical characteristics, on users' intentions to lurk. Furthermore, previous investigations into the lurking phenomenon have yielded conflicting viewpoints without identifying its underlying mechanism. Some researchers posit lurking as a positive behavior that should be encouraged (Takahashi *et al.*, 2002; Edelman *et al.*, 2017), while others perceive it negatively (Mo and Coulson, 2010). Interestingly, users sometimes choose to lurk due to fear of negative outcomes (Ortiz *et al.*, 2018) or discomfort in online social environments (Liu *et al.*, 2020). This ambiguity may arise from the dual nature of technology itself; technostress can be both positive and negative, possessing a "dark side" in addition to a "bright side" (Califf *et al.*, 2020; Breward *et al.*, 2017; Tarafdar *et al.*, 2014). The objective of this study is to conceptualize a technostress process that encompasses the negative components of technostress, which are embedded in users' maladaptive lurking behaviors. Currently, the impact of the SNS distinct social environment on users' maladaptive lurking intentions remains unclear, particularly concerning the integration of users' diverse social networks and the unique boundary-spanning media features.

SNS differ from traditional online communities in several distinctive features (Liu *et al.*, 2020). Primarily, SNS are designed to facilitate the creation of social networks and establish an individual's online presence, whereas traditional online communities primarily aim to enhance one's understanding of a specific topic (Zhu and Chang, 2014). Thus, users in an online community are likely to have similar goals, rules, and interests, and therefore share a common group identity, which refers to the common cognitive state of users (Sun *et al.*, 2014). On SNS, users form connections primarily because they are interested in the individual behind the profile and wish to maintain reciprocal relationships. Moreover, users' social networks on SNS are highly diverse, allowing them to assume as many identities as the distinct networks of relationships in which they participate and play various roles. From a technical perspective, SNS mix all of a user's social roles together, which can potentially impede their online presence and ultimately affect their level of online participation. As a result, the motivations for users to visit and post on SNS differ from those in traditional online communities (Zhu and Chang, 2014). This also implies that the mechanism of lurking on SNS should differ from those in traditional online communities. To fill this research gap, this paper aims to explore the mechanism of how the social environment of SNS influences users' lurking intentions.

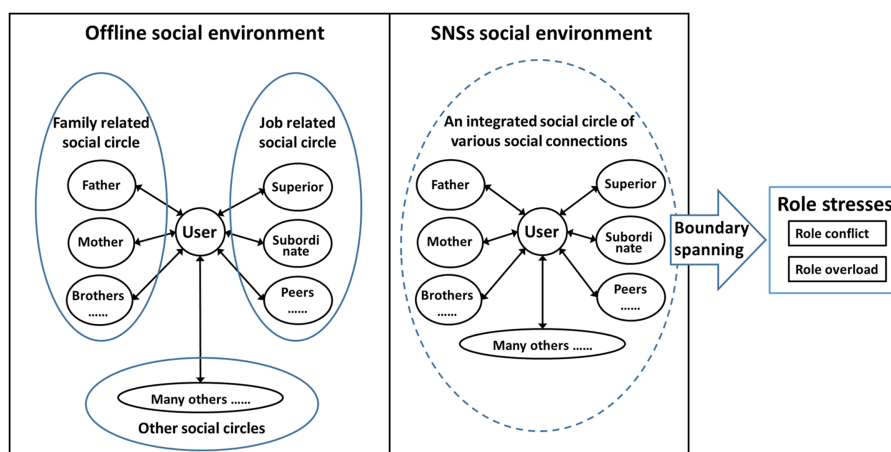
2.2 Role theory

Role theory (Kahn *et al.*, 1964), one of the earlier models focuses on the person's role at work as a major source of occupational stress. A role is generally defined as a person's perception of himself/herself when he/she relates to the surrounding environment and people whom he/she interacts with (Hall, 1971). As an example, an individual may serve as a father in a family, as a subordinate at work, or as a friend with some familiar people. Role stresses or demands are seen as coming from organizational and environmental factors, which may adversely

affect job satisfaction and health (Hall, 1971; Kahn *et al.*, 1964). The basic assertion is that social-psychological factors in the contemporary environment have major effects on the physical and psychological well-being of the person (Jr., 1968). This study aims to investigate how the unique context-specific cues of SNS can induce users' psychological burden from a role perspective, subsequently leading to lurking intention on these platforms. Therefore, role theory serves as a fitting theoretical paradigm to elucidate the mechanisms underlying this topic.

Given the distinctive social dynamics of SNS (Zhang *et al.*, 2011; Liu *et al.*, 2020), we employ a socio-technical perspective to analyze the context-specific cues. SNS are socio-technically constructed, where the social elements pertain to users' social networks, and the technical elements relate to the characteristics of the media. As depicted in Figure 1, the pervasive use of SNS in daily life has significantly expanded people's social networks. Many offline social connections are transitioning online, and new relationships are being formed. As the number of social connections on SNS grows, users' social circles become increasingly diverse, encompassing relatives, friends, acquaintances, coworkers and many others. Concurrently, SNS offer users a unique technical environment where all social circles coexist, thereby blurring the boundaries of their various social circles. In such circumstances, individuals often engage in frequent boundary-spanning activities, which can lead to issues with role pressure.

In this paper, we adopt a socio-technical perspective to address SNS context-specific cues, considering the size of the social network as a social cue and boundary spanning as a technical cue. Boundary spanning refers to interdepartmental and external communication; it involves having a large role set and requires interactions with many people, both inside and outside the organization, who have diverse needs and expectations (Lysonski, 1985; Goolsby, 1992). Specifically, we examine boundary spanning as a second-order construct, encompassing two first-order constructs: *parallelism* and *reprocessability*. Parallelism refers to the degree to which signals from multiple senders can be simultaneously transmitted over the medium (Dennis and Valacich, 2008). Media with high parallelism allow users to send and receive messages to and from multiple participants. High parallelism media allow users to send and receive messages to multiple participants (He and Yang, 2016; Tang *et al.*, 2013). This feature enables users to engage in social interactions regardless of their location. On the other hand, reprocessability is



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Figure 1.
Offline social
environment VS SNS
social environment

the extent to which the medium allows a message to be reexamined or processed again, either within the context of the communication event or after the event has concluded (Dennis and Valacich, 2008). Reprocessability enables users to access information about previous interactions, assisting new participants in understanding past activities (He and Yang, 2016). This feature allows users to engage in social interactions at any given time. Together, these two media capabilities form the unique technical environment of SNS. This environment liberates users' social interactions from the constraints of time and space, thereby transforming the structure of their social interactions (Zhang *et al.*, 2011).

Moreover, having a large social network size often results in diverse social circles, and the boundary-spanning technical features can blend these circles together. Consequently, users need to frequently engage in boundary-spanning activities to facilitate social interactions on SNS. When communicating across different social circles, users must adapt their behavior and roles to meet the unique criteria each circle uses to evaluate its members (Zhang *et al.*, 2011). Each role comes with its own set of expectations that users are expected to fulfill (Jr., 1968). However, these expectations can vary greatly and may even conflict between different roles (Biddle, 1979). This makes it challenging for users to navigate the frequent transitions between roles, potentially leading to discomfort and cognitive or emotional strain. The role transitions become more difficult when a person moves among more segmented roles (Ashforth *et al.*, 2000). Role conflict and role overload are two consequences of role transitions during these boundary-spanning activities on SNS (Zhang *et al.*, 2011).

Role conflict is defined as the incompatibility and incongruity in the expectations associated with a specific role (Guimaraes and Igbaria, 1992). As users engage in social conversations on SNS, they often conduct boundary-spanning activities across multiple roles. However, the expectations from each role are often inconsistent and conflicting. These intertwined role expectations can clash with each other and cannot be simultaneously fulfilled, thereby increasing the likelihood of role conflicts. For instance, if a doctoral student shares a travel photo on SNS, his friends might "like" it, whereas his supervisor might encourage him to focus more on completing his doctoral thesis.

Role overload refers to the scenario where the various role expectations placed on an individual exceed the resources they have available to manage them appropriately (Bostrom, 1980). Given the limited cognitive resources a user can process within a specific timeframe (Carpenter *et al.*, 1994), an increase in role overload can lead to individuals feeling overwhelmed by information processing (Ho *et al.*, 2003). For example, a user may be engaged in multiple conversations with diverse groups on SNS, assuming multiple roles simultaneously, such as a friend, spouse and colleague. Therefore, actively participating in all SNS conversations without making errors or adjustments can pose a significant challenge to the user.

Role conflict and role overload can arise from diverse social networks, leading role incumbents to inadequately manage their behaviors (Peterson, 1995). When social conversations involve conflicting roles on SNS, a user must navigate the challenge of satisfying one person's expectations while fulfilling another's. Moreover, when the number of overlapping roles becomes excessive, the social conversations can overwhelm the user. Consequently, role conflict and role overload can make it challenging for users to adjust their behaviors appropriately on SNS, causing them to exert more effort to meet others' expectations. This can lead to users feeling emotionally drained, a state referred to as SNS fatigue (Bright *et al.*, 2015; Dhir *et al.*, 2018; Karr-Wisniewski and Ying, 2010). Role theory introduces a new perspective to explain how SNS contextual-specific cues lead to users' perceptions of role stresses, but cannot adequately explain users' lurking intention because of its weakness in elaborating on users' motivational behavior (Miles, 2012). Thus, burnout theory serves as another theoretical basis for this study to explain the psychological and behavioral responses of role stresses, which we described as follows.

2.3 Burnout theory

Burnout theory has been especially useful in uncovering the role of work-specific role stressors in predicting job burnout and satisfaction (Bacharach *et al.*, 1991; Maslach and Jackson, 1981; Maslach and Goldberg, 1998; Jackson *et al.*, 1987; Jackson and Schuler, 1985). The theory suggests that burnout occurs when certain valued resources are lost, are inadequate to meet demands, or do not yield the anticipated returns. Certain behavioral and attitudinal outcomes are likely to occur as a result of resource loss and burnout. The major outcomes include behavioral coping responses; turnover intentions; and erosion of organizational commitment, job involvement and job satisfaction (Kahill, 1988). Specifically, Maslach *et al.* (1996) provided a general model of burnout, with major antecedents and consequences, which stated that work overload and personal conflict are the major antecedents of burnout, and the consequences of burnout are seen most consistently in various forms of job withdrawal (i.e. decreased commitment, job dissatisfaction, turnover and absenteeism). This model introduces a theoretical framework that sheds light on how role conflict and role overload lead to significant exhaustion among SNS users, which in turn, leading to their lurking intentions.

According to Maslach and Jackson (1981), burnout consists of three dimensions: emotional exhaustion, depersonalization and diminished personal accomplishment. Emotional exhaustion as a form of strain is directly affected by role conflict and quantitative workload, whereas depersonalization as a form of defensive coping and personal accomplishment as a form of self-evaluation are indirectly affected by emotional exhaustion (Jackson *et al.*, 1987). Thus, this study employed the concept of SNS fatigue to depict users' exhausted emotions on SNS, which has been adapted to the field of information systems research (Dhir *et al.*, 2018; Zhang *et al.*, 2016; Ravindran *et al.*, 2014; Fan *et al.*, 2021). SNS fatigue refers to a subjective, self-evaluated feeling of exhaustion stemming from the use of SNS (Zhang *et al.*, 2016; Ravindran *et al.*, 2014; Lee *et al.*, 2016; Fan *et al.*, 2021). Existing literature primarily explores the precursors of SNS fatigue, focusing on aspects such as self-efficacy, privacy concerns and perceived usefulness (Bright *et al.*, 2015), system feature overload, information overload and social overload (Zhang *et al.*, 2016), as well as compulsive use and fear of missing out (Dhir *et al.*, 2018), among others. However, the existing literature exhibits a pronounced gap regarding the exploration of the causes and consequences of SNS fatigue, especially from the perspective of the various roles embedded within users' social networks.

Interactions within the social environment of SNS significantly influence users' behavior on these platforms. Role conflict and role overload can pose challenges for users, making it difficult for them to adjust their behaviors on SNS appropriately. This can lead to negative evaluations from others and a failure to maintain their online relationships and images (Liu *et al.*, 2020). The cognitive stress from managing multiple roles can consume significant time and energy, leading to emotional fatigue among SNS users. Additionally, experiences of fatigue may ultimately drive a user to withdraw from a stressful environment, such as by reducing or suspending activity (Ravindran *et al.*, 2014). Lurking thus emerges as a prevalent form of withdrawal for SNS users when dealing with such circumstances. In this context, we specifically chose lurking intention as a proxy for lurking behavior, given that lurking is a covert behavior that is challenging to observe. Furthermore, in the field of information systems, the notion that behavioral intentions determine behavior is widely recognized (Ajzen, 1985). As per the lurking literature review mentioned earlier, lurking intention is defined as the tendency to reduce or even cease posting content on SNS.

3. Research model and hypotheses

Based on the role theory and burnout theoretical model, this study aims to investigate the impact of SNS context-specific cues on users' lurking intentions. Our proposed research model, as depicted in Figure 2, suggests a general framework. It posits that the social

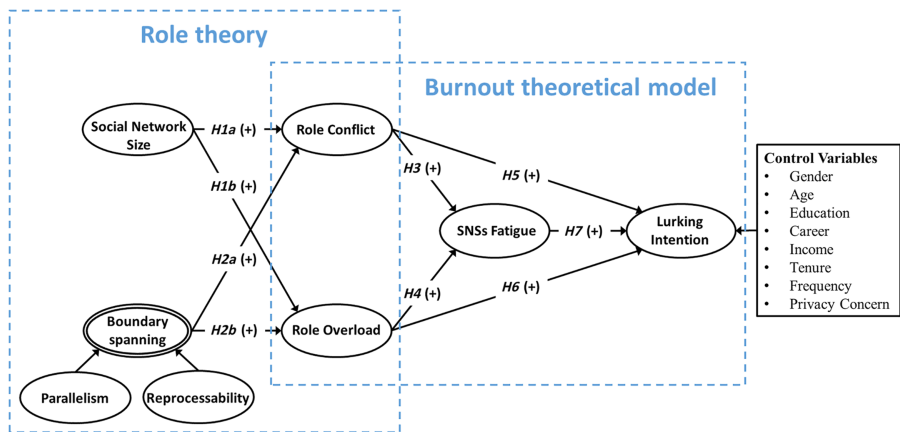


Figure 2.
The research model

Note(s): Single-line ellipses represent first-order constructs; double-line ellipses represent second-order constructs
Source(s): Authors' own creation

network size and boundary spanning act as context-specific cues, leading to role conflict and role overload. These, in turn, contribute to user fatigue on SNS. Both role stresses and SNS fatigue subsequently impact users' lurking intentions.

3.1 The influence of SNS context-specific cues on role stresses

As SNS continue to rise in popularity, users are increasingly shifting their social connections from offline to online platforms, while also forming new connections. This shift has led to a continuous expansion of their social network size. For example, the average number of friends per WeChat user is approximately 128, a figure that is expected to grow as more users join the workforce (Tencent, 2015). WeChat has recently expanded the maximum number of contacts a user can have from 5,000 to 10,000. This enhancement reflects the platform's commitment to adapting its network capabilities to meet the growing demands and needs of its users. As the number of contacts increases, the diversity within users' social circles on SNS also grows. Notably, WeChat has become a hub for most people's social circles, encompassing relatives, friends, supervisors, colleagues, customers and many others. People need to take on different roles in different social environments (social circles), and each role is associated with a different set of role expectations (Zhang et al., 2011). Role expectations are frequently inconsistent and excessive, with the demands placed on the individual exceeding the available resources to manage them effectively. Thus, this interplay of diverse role expectations can lead to role stresses (Zhang et al., 2011; Liu and Wang, 2018). One form of this stress is role conflict, which arises from the incompatibility between different role expectations and the challenges in reconciling them (Guimaraes and Igbaria, 1992). Another form is role overload, which is caused by an excess of role expectations (Bostrom, 1980). Given these observations, the following hypotheses are proposed:

H1a. Social network size positively affects users' perceived role conflict on SNS.

H1b. Social network size positively affects users' perceived role overload on SNS.

SNS offer users a unique environment that spans across various social boundaries, bringing together diverse social circles (Hampton et al., 2011). This allows users to engage in social

conversations without the limitations of time and space, as they can access these different social circles anytime, anywhere (Zhang *et al.*, 2011; Liu *et al.*, 2020). However, this also necessitates the assumption of multiple roles simultaneously, leading to frequent boundary-spanning activities. When people must conduct cross-boundary activities, they need to switch their roles accordingly. The role transitions become more difficult when a person moves among more segmented roles (Ashforth *et al.*, 2000). Role conflict and role overload are two consequences of role transitions (Friday and Hage, 1976; Zhang *et al.*, 2011; Liu *et al.*, 2020). When individuals engage with a social group, they typically aim to act and assess themselves according to the group's norms. However, these norms often vary among different social groups. When social activities span multiple groups with divergent norms, it becomes challenging for an individual to simultaneously meet these conflicting expectations. Consequently, role transitions resulting from boundary-spanning activities emerge as the primary source of role conflict (Baroudi, 1985; Lysonski, 1985). Moreover, due to cognitive constraints, an increase in boundary-spanning activities results in having limited time and cognitive resources to meet these various role expectations (Ho *et al.*, 2003; Zhang *et al.*, 2011). Thus, the overwhelming number of roles to manage can make it seem impossible for users to meet all their role expectations within a constrained timeframe and energy capacity, ultimately leading to a sense of role overload. Therefore, the following hypotheses are proposed:

H2a. Boundary spanning positively affects users' perceived role conflict on SNS.

H2b. Boundary spanning positively affects users' perceived role overload on SNS.

3.2 The influence of role stresses on SNS fatigue

The social environment of SNS increases the likelihood of users engaging in multiple open conversations (Zhang *et al.*, 2011). However, when these conversations involve conflicting roles, they overwhelm the users faster. Role conflict arises when the expectations from different roles are contradictory and cannot be simultaneously fulfilled. To maintain relationships and establish a positive online image, users often need to invest significant time and energy to balance these conflicting behaviors. For instance, a user may be involved in multiple social conversations with different groups on SNS at the same time. Actively engaging in all these conversations without making errors or adjustments can pose a real challenge. Consequently, users who frequently find themselves in situations of role conflict may feel emotionally drained due to excessive energy consumption and anxiety over their ability to manage their social conversations properly (Xiao *et al.*, 2022; Fan *et al.*, 2021). This leads to the following hypothesis:

H3. Role conflict positively affects users' perceived SNS fatigue.

When interacting with various social circles, individuals often find themselves needing to switch between different roles, each with its own set of expectations (Ashforth *et al.*, 2000). This is particularly true on SNS, where a user can be engaged in multiple conversations with diverse groups of people simultaneously (Zhang *et al.*, 2011). Given that people have a limited amount of cognitive resources available to process information within a specific timeframe (Carpenter *et al.*, 1994), an increase in role overload can result in individuals feeling overwhelmed (Ho *et al.*, 2003). This is because role overload consumes a significant portion of a user's limited cognitive resources, making it challenging for them to cope (Ho *et al.*, 2003). Consequently, this can lead to a sense of fatigue associated with the use of SNS. This leads to the following hypothesis:

H4. Role overload positively affects users' perceived SNS fatigue.

3.3 The influence of role stresses and SNS fatigue on lurking intention

In the context of role conflict on SNS, users often find themselves juggling two or more conflicting activities. This makes it difficult for them to manage and control information during social interactions (Zhang *et al.*, 2011). At times, they may need to violate a rule or norm to accomplish their desired tasks, or they may feel compelled to suppress their genuine feelings and thoughts to reconcile the conflicting activities (Rutner *et al.*, 2008). If their actions are approved by some but disapproved by others, they may experience anxiety due to the fear of anticipated sanctions or negative reactions from others (Strauman and Higgins, 1987; Levinson and Rodebaugh, 2013). Individuals prone to anxiety often avoid social interactions when they are linked with stress (Kashdan, 2007) and they resort to safety behaviors to mitigate their anxiety (Rachman *et al.*, 2008). Consequently, out of fear of miscommunication in social interactions, anxious users tend to limit self-disclosure to prevent negative social outcomes, which is a part of their safety behaviors (Rapee and Heimberg, 1997; Hong *et al.*, 2015). As a result, lurking becomes a safer alternative for users on SNS, allowing them to avoid active participation in conversations (Osatuyi, 2015). This leads to the following hypothesis:

H5. Role conflict positively affects users' lurking intention on SNS.

Users typically engage with a variety of social circles on SNS, each of which comes with its own set of roles and expectations. However, when the number of roles from these multiple networks becomes overwhelming, users may experience role overload (Zhang *et al.*, 2011). This is due to cognitive limitations that restrict the amount of information users can process within a given timeframe (Carpenter *et al.*, 1994). To manage these excessive role expectations, users are compelled to invest more efforts and resources into their SNS interactions. This significant investment of time and energy can adversely affect their daily lives, potentially causing disruptions to their work and other important activities (Zhang *et al.*, 2016; Ravindran *et al.*, 2014). To mitigate these disruptions and negative impacts, users may need to distance themselves from excessive social interaction. In this context, lurking becomes a viable solution for users to alleviate the stress associated with role overload on SNS. This leads us to the following hypothesis:

H6. Role overload positively affects users' lurking intention on SNS.

SNS fatigue is a subjective and negative sensation characterized by exhaustion, boredom and burnout, which stems from excessive SNS activities (Zhang *et al.*, 2016). This is particularly prevalent when users' social interactions on SNS frequently involve role stresses. To mitigate the physical and psychological strain induced by such negative emotions, individuals often attempt to alter their circumstances (Lavee *et al.*, 1987). This could involve discontinuing stress-inducing behaviors or adopting social avoidance strategies. Lurking, a form of social avoidance strategy, can diminish users' exposure to social activities, thereby circumventing the additional social efforts and cognitive resources required to manage SNS fatigue. Users of SNS are progressively "de-socializing" to protect themselves from the negative emotional responses triggered by excessive social activities. Consequently, their usage motivation is gradually shifting toward a singular direction of merely consuming social content. This observation leads to the following hypothesis:

H7. SNS fatigue positively affects users' lurking intention on SNS.

3.4 Control variables

Demographic characteristics of individuals, such as gender, age, education, occupation and income, along with their experiences using SNS, including the number of friends, tenure and frequency of use, have been frequently suggested to influence SNS usage (Zhang *et al.*, 2016;

Lankton *et al.*, 2017). Furthermore, the impact of privacy concerns on SNS posting behaviors has been confirmed in previous studies (Everard *et al.*, 2011; Lankton *et al.*, 2017; Mousavi *et al.*, 2020). Notably, privacy concerns have also been identified as a significant influencing factor for lurking (Sun *et al.*, 2014; Ortiz *et al.*, 2018). Consequently, all these elements have been integrated into our research model as control variables, specifically in relation to the proposed lurking intention (refer to Figure 2).

4. Methodology

4.1 Instruments

The measurement tools for this study were theoretically validated by prior research, and we adapted them for our purposes. As the original items were in English, the instruments were first translated into Chinese and then back-translated (i.e. from Chinese to English) (Zhang *et al.*, 2016). This process was performed by three professors proficient in both languages. Any discrepancies or inconsistencies in the translation process were resolved to ensure the final instruments were of high quality. Before the formal survey was launched, a focus group was assembled. This group consisted of three researchers from the field of information systems and five graduate students who were experienced users of SNS. Their task was to further verify the measurement items. Additionally, we invited 15 SNS users to participate in a pilot study to refine the clarity of instructions and questions in the online questionnaire. To gather more detailed feedback, the respondents were invited for face-to-face interviews to share their thoughts on the questionnaire. After analyzing the feedback, we made several minor revisions to refine the questionnaire. The final scales are included in Appendix 2. All items were measured using seven-point Likert scales ranging from “strongly disagree” (1) to “strongly agree” (7).

4.2 Data collection

Participants for this study were recruited from the user base of WeChat, a cross-platform instant messaging service developed by Tencent on January 21, 2011. WeChat shares similar features with WhatsApp, allowing users to send both text and voice messages. A key feature of WeChat is WM, which serves as the primary social networking function within the app. WM is the leading SNS in China, boasting a vast user base and immense popularity. Unlike other SNS platforms such as Microblog, Facebook and Twitter, WM is primarily used for communication among relatives, friends, acquaintances and other social connections of the users. This makes it an ideal platform for studying the effects of SNS context-specific cues from a role perspective. An online random survey was conducted on Sojump, a professional survey distribution platform in China from May 23 to June 2, 2017. Respondents who successfully completed and submitted the survey were given a 5 RMB monetary reward. In total, 692 WM users from mainland China completed our questionnaire. Because of the length of the survey, and to decrease monomethod bias, we followed the best practice of using two attention-trap questions to determine if the respondents had read all the questions fully and answered honestly (Lowry *et al.*, 2013). After removing invalid responses, such as those failing our attention-trap questions and those providing identical answers across all questions, we were left with a total of 491 valid responses (71% valid response rate). We assessed non-response bias by comparing the means of all variables and demographics between early and late respondents. The *t*-test results showed no significant differences. As shown in Table 1, about 44.81% of the respondents were male, and most respondents were aged between 19 and 39 years (84.52%). In addition, 99.19% of users had used WM for more than one year. On average, 96.95% of users checked their mobile SNSs several times a day. This demographic profile aligns well with the typical user base of WeChat.

Demographic characteristics	Measure	Frequency	Percent (%)
Gender	Male	220	44.81
	Female	271	55.19
Age	≤19	13	2.65
	≤29	281	57.23
	≤39	134	27.29
	≤49	55	11.20
	≥50	8	1.63
	High school or below	3	0.61
Education	Associate degree	26	5.30
	Bachelor's degree	200	40.73
	Master's degree	195	39.71
	Doctoral degree or higher	67	13.65
	Student	193	39.31
	Civil servant	6	1.22
Career	Institution staff	68	13.85
	Enterprise staff	181	36.86
	Self-employed	6	1.22
	Others	37	7.54
	≤2000	115	23.42
	≤5,000	59	12.02
Income (Yuan/monthly)	≤8,000	111	22.61
	≤15,000	74	15.07
	>15,000	132	26.88
	≤50	40	8.15
	≤100	75	15.28
	≤150	66	13.44
Number of friends	≤200	83	16.90
	≤250	55	11.20
	≤300	49	9.98
	≤350	31	6.31
	>350	92	18.74
Tenure (years since registered)	≤1	4	0.81
	≤2	28	5.71
	≤3	102	20.77
	≤4	119	24.24
	≤5	84	17.11
	>5	154	31.36
Frequency	Hourly	322	65.58
	Several times a day	154	31.37
	Several times a week	13	2.65
	Several times a month	1	0.20
	less or none	1	0.20
Total		491	100

Table 1.
Demographic
characteristics of the
respondents

Source(s): Authors' own creation

5. Results

The model and hypotheses testing were conducted using SmartPLS 4.0. Partial Least Squares Structural Equation Modeling (PLS-SEM) has gained popularity in contemporary research due to its distinct advantages, such as minimal requirements on measurement scales, sample distribution and sample size. The primary goal of PLS-SEM analyses is to maximize the variance explanation of a model's endogenous constructs (Hair *et al.*, 2011). It is particularly effective in causal-predictive analysis, especially when the hypothesized relationships are complex and have few established bases (Zhang *et al.*, 2018). We selected PLS-SEM for its

ability to facilitate the discovery-oriented or theory development process in research aimed at identifying the key drivers of a construct. It is also capable of modeling complex latent constructs with a large number of items (Hair *et al.*, 2019; Shiau *et al.*, 2019). In a research model that only includes reflective constructs, the sample size required by PLS-SEM should be at least 10 times the larger number of paths leading to an endogenous construct (Chin, 1998). In our research model, all latent constructs were implemented as reflective, and the maximum number of paths entering an endogenous construct was 11. Therefore, our sample size of 491 was sufficient for analyzing our research model using the PLS-SEM technique.

5.1 Common method bias

In our study, we acknowledged the potential for common method biases that could arise from various sources, such as consistency motif and social desirability, as is the case with all self-reported data (Podsakoff and Organ, 1986; Philip *et al.*, 2003). To evaluate the extent of this common method bias, we conducted statistical analyses. Initially, we performed a Harmon one-factor test, as suggested by Podsakoff and Organ (1986). This test was applied to seven key variables in our theoretical model: parallelism, reprocessability, role conflict, role overload, SNS fatigue, lurking intention and privacy concern. The results showed that the maximum covariance explained by a single factor was 29.064%. This indicates that no single factor could account for the majority of the variance, suggesting that our results are unlikely to be significantly affected by common method biases. Furthermore, we considered the measured latent marker-variable approach, which is known to help detect and control common method bias (Malhotra *et al.*, 2006). To implement this marker-variable technique, we carefully selected variables that were unrelated to the phenomena being investigated. As such, we chose “fashion consciousness,” defined as the importance an individual places on being fashionably dressed (Lumpkin and Darden, 1982). The results demonstrated that the path coefficients and model fit values were in line with the original estimates (as shown in Table 2). Consequently, we concluded that common method bias does not pose a significant concern to our overall study.

5.2 Assessment of the measurement model

The measurement model was evaluated to confirm item reliability, convergent validity, and discriminant validity. A common method for verifying individual item reliability is to check if individual item loadings exceed 0.60, or ideally, 0.70 (Barclay *et al.*, 1995). In this study, the measurement items in the model generally loaded heavily on their respective constructs, as

Relationship	Without marker		With marker	
	Path coefficient	P-value	Path coefficient	P-value
Social Network Size → Role Conflict	0.098*	0.029	0.097*	0.031
Social Network Size → Role Overload	0.106*	0.020	0.100*	0.027
Boundary Spanning → Role Conflict	0.116*	0.019	0.113*	0.021
Boundary Spanning → Role Overload	0.132*	0.012	0.119*	0.020
Role Conflict → SNS Fatigue	0.229***	0	0.232***	0
Role Overload → SNS Fatigue	0.389***	0	0.383***	0
Role Conflict → Lurking Intention	0.100*	0.048	0.120*	0.029
Role Overload → Lurking Intention	0.111*	0.034	0.127*	0.018
SNS Fatigue → Lurking Intention	0.409***	0	0.449***	0

Note(s): * $p < 0.05$; *** $p < 0.001$

Source(s): Authors' own creation

Table 2.
Measured latent
marker-variable
approach for common
method bias

shown in Table 3, with loadings surpassing 0.6, thereby demonstrating satisfactory reliability. Table 4 reveals that the Cronbach's α and composite reliability of each construct exceeded the recommended value of 0.8. Furthermore, the average variance extracted (AVE) for all constructs was above 0.5, and the rho_A values were >0.7 (Henseler *et al.*, 2016). This indicates that the criteria for internal consistency were met. The third step in evaluating the

Table 3.
Matrix of cross-loadings

	Par	Rep	RC	RO	SNSF	LI	PC
Par1	0.896	0.642	0.078	0.112	0.002	0.035	0.164
Par2	0.846	0.532	0.104	0.080	−0.043	0.002	0.101
Par3	0.899	0.635	0.142	0.120	0.011	0.060	0.160
Par4	0.853	0.641	0.057	0.085	−0.036	0.045	0.161
ReP1	0.653	0.901	0.089	0.115	0.026	0.049	0.160
ReP2	0.646	0.907	0.081	0.135	0.051	0.078	0.206
ReP3	0.540	0.823	0.157	0.158	0.088	0.085	0.154
ReP4	0.575	0.809	0.098	0.104	0.043	0.101	0.123
RC1	0.035	0.056	0.772	0.448	0.327	0.304	0.109
RC2	0.053	0.050	0.765	0.517	0.328	0.277	0.215
RC3	0.166	0.163	0.831	0.584	0.391	0.333	0.253
RC4	0.083	0.106	0.849	0.576	0.491	0.417	0.256
RO1	0.180	0.235	0.551	0.649	0.288	0.218	0.168
RO2	−0.007	0.046	0.553	0.800	0.411	0.323	0.247
RO3	0.087	0.086	0.550	0.788	0.395	0.344	0.237
RO4	0.217	0.215	0.450	0.728	0.408	0.357	0.248
RO5	0.034	0.061	0.512	0.860	0.483	0.391	0.230
RO6	0.050	0.085	0.515	0.823	0.502	0.412	0.256
SNSF1	0.117	0.158	0.368	0.382	0.735	0.389	0.245
SNSF2	−0.028	0.052	0.467	0.544	0.921	0.567	0.256
SNSF3	−0.044	0.007	0.415	0.461	0.902	0.507	0.281
SNSF4	−0.083	0.004	0.410	0.447	0.847	0.472	0.254
LI1	0.086	0.106	0.380	0.396	0.507	0.910	0.331
LI2	0.051	0.071	0.347	0.393	0.471	0.917	0.299
LI3	0.119	0.155	0.371	0.398	0.480	0.916	0.331
LI4	−0.116	−0.026	0.362	0.362	0.526	0.725	0.294
PC1	0.098	0.119	0.260	0.269	0.249	0.330	0.817
PC2	0.153	0.155	0.199	0.225	0.258	0.287	0.860
PC3	0.150	0.192	0.230	0.273	0.283	0.326	0.899
PC4	0.186	0.186	0.224	0.264	0.254	0.307	0.886

Note(s): Parallelism (Par); Reprocessability (Rep); Role conflict (RC); Role overload (RO); SNS fatigue (SNSF); Lurking intention (LI); Privacy concern (PC)
Source(s): Authors' own creation

Table 4.
Reliability test

Construct	AVE	CR	Cronbach's α	rho_A
Parallelism	0.764	0.928	0.897	0.899
Reprocessability	0.742	0.920	0.883	0.888
Role Conflict	0.648	0.880	0.821	0.842
Role Overload	0.605	0.901	0.868	0.879
SNS Fatigue	0.730	0.915	0.874	0.893
Lurking Intention	0.758	0.926	0.890	0.891
Privacy Concern	0.750	0.923	0.889	0.890

Note(s): Average variance extracted (AVE); Composite reliability (CR)
Source(s): Authors' own creation

measurement model involved assessing its discriminant validity. As per Table 5, the Fornell and Larcker criterion suggests that the square roots of the AVEs (the bold numbers in the diagonal) were greater than the correlations between the latent variable and other latent variables (Fornell and Larcker, 1981). Table 6 shows that the values of the heterotrait–monotrait ratio of correlations were below the threshold of 0.9 (Henseler *et al.*, 2016). These results provide ample evidence of discriminant validity for these constructs. Discriminant validity is further substantiated when the loadings for the items on their targeted constructs are higher than loadings on other constructs in the model. Table 3 includes the loadings and cross-loadings for items used in this study, and it is evident that all items load more heavily on their constructs than they do on any other constructs. We also used the variance inflation factor (VIF) to test multicollinearity (see Appendix 3). The VIFs are lower than the threshold of 5.0 (Yu *et al.*, 2023), indicating the absence of multicollinearity.

Boundary spanning was conceptualized as a second-order formative construct, composed of two first-order components. To test these second-order formative constructs, we employed PLS-SEM. As suggested by Edwards (2001) and Chin *et al.* (2003), the paths from the lower-order to the higher-order construct should be initially modeled. In the first step of our analysis, we used the scores of the latent variables from the PLS-SEM as formative measures for the second-order constructs. As depicted in Figure 3, the second-order formative models represent the relationship between the first-order and the second-order factors in PLS-SEM. All the weights from the first-order components to the second-order constructs were highly significant ($p < 0.001$). However, multicollinearity can pose a problem for measures of formative constructs (Jarvis *et al.*, 2003). It is suggested that formative measures should not exhibit strong correlations, as high multicollinearity can destabilize the construct. Moreover, high correlations may indicate that multiple measures are tapping into the same dimension of the construct. In light of this, our study also examined potential multicollinearity among the first-order components by conducting a variance inflation factor (VIF) test. As shown in

	Par	Rep	RC	RO	SNSF	LI	PC
Parallelism (Par)	<i>0.874</i>						
Reprocessability (Rep)	0.703	<i>0.861</i>					
Role Conflict (RC)	0.109	0.121	<i>0.805</i>				
Role Overload (RO)	0.114	0.148	0.664	<i>0.778</i>			
SNS Fatigue (SNSF)	−0.018	0.059	0.488	0.541	<i>0.854</i>		
Lurking Intention (LI)	0.041	0.09	0.421	0.446	0.572	<i>0.871</i>	
Privacy Concern (PC)	0.169	0.188	0.265	0.299	0.302	0.362	<i>0.866</i>

Note(s): The italic fonts in the leading diagonals are the square root of AVEs; Off-diagonal elements are correlations among latent constructs

Source(s): Authors' own creation

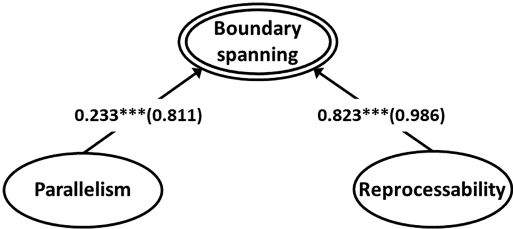
Table 5.
Discriminant validity:
Fornell-Larcker
criterion

	Parallelism	Rep	RC	RO	SNSF	LI
Reprocessability (Rep)	0.787					
Role Conflict (RC)	0.125	0.139				
Role Overload (RO)	0.142	0.179	0.79			
SNS Fatigue (SNSF)	0.094	0.084	0.561	0.609		
Lurking Intention (LI)	0.120	0.121	0.482	0.500	0.644	
Privacy Concern (PC)	0.189	0.211	0.301	0.337	0.344	0.406

Source(s): Authors' own creation

Table 6.
Discriminant validity:
Heterotrait-Monotrait
ratio of correlations

Figure 3.
Measurement of
second-order formative
construct



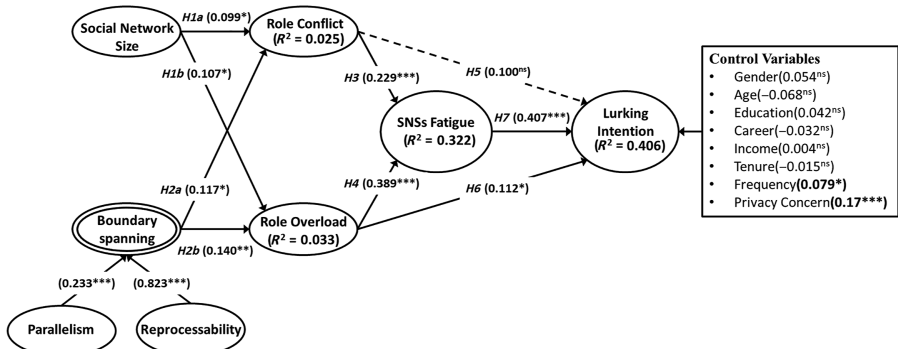
Note(s): *** $p < 0.001$; n1 (n2): n1 represents weight, n2 represents loading
Source(s): Authors' own creation

Table 7.
Second-order
formative construct
variance inflation
factor (VIF)

Second-order construct	First-order construct	VIF
Boundary spanning	Parallelism	1.976
	Reprocessability	1.976

Source(s): Authors' own creation

Figure 4.
Results of hypotheses
testing



Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, ns non-significant
Source(s): Authors' own creation

indicating that an increase in usage frequency heightened users' lurking intention on SNS. This could be attributed to the fact that as users gain more experience, they encounter more stimuli and experience role stresses on SNS, thereby reinforcing their lurking intention. Furthermore, privacy concerns significantly and positively influenced lurking intention, aligning with previous research (Sun *et al.*, 2014; Ortiz *et al.*, 2018).

5.4 Overall model fit evaluation

We assessed the research model's goodness of fit by scrutinizing the standardized root mean square residual (SRMR), unweighted least squares discrepancy (dULS) and geodesic discrepancy (dG) values (Henseler *et al.*, 2016). The lower these values (SRMR, dULS and dG), the more accurately the theoretical model fits (Henseler *et al.*, 2016). The results of the model fit are presented in Table 8. The SRMR value was beneath the suggested threshold of 0.08, and all discrepancies fell below the 95% bootstrap quantile (HI95) (Henseler *et al.*, 2016). This indicates a strong alignment between the research model and the collected data.

5.5 Robustness test

Two additional tests were conducted to check for the robustness of the proposed research model. First, social network size was treated as a dichotomous variable (low:0 vs high:1). Specifically, to ensure that the sample sizes of the two groups were as balanced as possible, users with a social network size of <200 were assigned to the lower group ($N = 264$), while the remaining users were placed in the higher group ($N = 227$). The results of the model test, as detailed in Appendix 4, were consistent with the findings from the hypothesis testing presented in Figure 4. Second, by considering users with a large social network size who infrequently use the SNS may exhibit different behaviors compared to those with a smaller social network size but more frequent usage. A bootstrap multigroup analysis was conducted to assess the differences between users who frequently (hourly, $N = 321$) access SNS and those who access less frequently ($N = 170$). As indicated in Appendix 4, users with a high frequency of SNS usage and large social network size perceive a greater intensity of role overload. Furthermore, the frequency of SNS usage exacerbates the relationship between role conflict and lurking intention, while simultaneously undermining the relationship between role overload and lurking intention. Thus, the results confirm the robustness of the proposed theoretical model and provide evidence that user characteristics significantly impact the lurking mechanism.

5.6 Post-hoc test of mediating effects

We conducted post-hoc tests to verify the validity of the model, specifically examining whether and how SNS fatigue mediates the relationships between role stresses (role conflict and role overload) and lurking intention. We followed the bootstrapping method recommended by Hayes (2017) to construct the confidence intervals (CIs) of the mediation effects. This allowed us to assess whether the direct effects of the independent variable (IV) on the dependent variable (DV) persisted after accounting for the effects of the mediators in

	Value	HI95	Conclusion
SRMR	0.041	0.046	Support
dULS	0.884	1.115	Support
dG	0.326	0.365	Support

Source(s): Authors' own creation

Table 8.
Model fit evaluation

the PLS-SEM. We calculated bias-corrected and accelerated CIs across 5,000 bootstrap resamples and set a 95% CI. As depicted in Table 9, the results suggested that SNS fatigue fully mediated the relationship between role conflict and lurking intention, and partially mediated the relationship between role overload and lurking intention. These findings highlight a distinction in the effects of role stresses on lurking intention. They also lend support to our proposed Hypothesis 5 (H5). Specifically, role conflict can indeed amplify users' lurking intention on SNS, but this effect is primarily channeled through SNS fatigue.

6. Discussion

6.1 Key findings

User-generated content is vital for the prosperity of SNS, as it generates new knowledge and value for various sectors including business (Lin *et al.*, 2021), finance (Schniederjans *et al.*, 2013; Asur and Huberman, 2010), decision-making (Kellner *et al.*, 2023; Sadovykh *et al.*, 2015) and public politics (Tumasjan *et al.*, 2010). However, user behavior on SNS often adheres to Pareto's law (Matei and Bruno, 2015), with over 80% of the content being created by less than 20% of active users. This study aimed to explore the impact of context-specific cues on users' lurking intentions on SNS. Our findings reveal that the size of a user's social network and the boundary-spanning technical feature can induce perceived role stresses, such as role conflict and role overload. These role stresses can significantly heighten users' SNS fatigue, which in turn, amplifies their lurking intention. Furthermore, SNS fatigue fully mediates the relationship between role conflict and lurking intention, and partially mediates the relationship between role overload and lurking intention.

Contrary to previous studies, which largely suggested that users with numerous social connections are more likely to post (Goes *et al.*, 2014; Stieglitz and Linh, 2013), our research presents a paradoxical finding. We discovered that a large number of social connections may actually decrease a user's intention to post on SNS due to the overwhelming role stresses associated with managing such networks. SNS are designed to provide users with a robust capability to reach out to others, manage their social relationships and control their online impressions (Chen *et al.*, 2016). Thus, most prior studies have focused on the positive aspects of these context-specific cues. For example, theories such as social influence (Posey *et al.*, 2010; Venkatesh *et al.*, 2003; Aral and Walker, 2014) and social capital (Wasko and Faraj, 2005; Kankanhalli *et al.*, 2005) suggest that large social networks provide users with sufficient incentives and potential social capital, thereby encouraging them to contribute more content. The Media Synchronicity Theory also identified five capabilities of media (symbol sets, rehearsability, transmission velocity, parallelism and reprocessability) that enhance media effectiveness (Tang and Wang, 2011; Dennis and Valacich, 2008). However, these studies often consider the effects of these context-specific cues in isolation, thereby overlooking the overall effects when these cues are combined. Our research offers a fresh perspective and additional evidence to explain the collective effects of these context-specific

Table 9.
Bootstrapping results
of mediation testing

Relationship	Mediating effect			Direct effect			Type of mediation
	2.5% lower bound	97.5% upper bound	Zero included?	2.5% lower bound	97.5% upper bound	Zero included?	
RC → SNSF → LI	0.045	0.149	No	−0.002	0.199	Yes	Fully
RO → SNSF → LI	0.108*	0.220	No	0.010	0.212	No	Partially

Note(s): Role conflict (RC); Role overload (RO); SNS fatigue (SNSF); Lurking intention (LI)
Source(s): Authors' own creation

cues. In particular, we identified boundary spanning as a second-order formative construct, composed of two dimensions of media capabilities (i.e. parallelism and reprocessability). This accounts for the boundary-spanning social environment of SNS, providing a new measurement and perspective for future studies to explore the negative aspects of SNS features.

Our results have confirmed that both role conflict and role overload can increase users' perceived fatigue from SNS, leading to an increased intention to lurk. [Fox and Moreland \(2015\)](#) suggested that IT features on SNS could have negative effects, causing individuals to experience fatigue and cognitive overload, as new technologies can influence risks, stress and illegal use. Previous studies have delved into the negative aspects of SNS, focusing on the mechanisms behind users' discontinuous usage intention ([Fan et al., 2021](#)) or their intention to stop using the platforms ([Dang, 2020](#)). Despite these issues, SNS remain crucial channels for people to access information and entertainment, thanks to their consistently high number of active users. It's important to note that users may reduce or even cease their content contribution on SNS for various reasons, but they continue to read and follow new messages. Therefore, lurking could be a more relevant research topic that warrants further attention. Current research on SNS lurking is limited and primarily contrasts the differences between lurkers and contributors ([Morrison et al., 2013](#); [Muller, 2012](#); [Tan, 2011](#); [Zheng et al., 2021](#)). The concept of lurking, as a passive usage behavior or social strategy, has been largely overlooked. Our study addresses this research gap, offering new theoretical and practical insights to better understand the lurking phenomenon on SNS.

6.2 Theoretical contributions

Given the limited research to date on lurking within SNS, this study makes three major contributions to our understanding of the evolution of SNS users' lurking intention. First, we delved into the context-specific cues of SNS from both social and technical perspectives. Specifically, we proposed that the distinctive social environment of SNS is shaped through the confluence of social network size and boundary-spanning technical features. While previous research has predominantly highlighted the positive implications of these context-specific factors on user behaviors, our study offers a fresh perspective, shedding light on their potential negative effects. Previous studies have extensively examined the positive impact of these two context-specific factors on users' active adoption behaviors, using various theoretical lenses such as social influence theory ([Posey et al., 2010](#); [Venkatesh et al., 2003](#); [Aral and Walker, 2014](#)), social capital theory ([Wasko and Faraj, 2005](#); [Kankanhalli et al., 2005](#)) and media synchronicity theory ([Tang and Wang, 2011](#); [Dennis and Valacich, 2008](#)). Yet, these frameworks tend to address these context-specific cues in isolation. In contrast, our study offers an integrated viewpoint, evaluating the influence of both cues on users' passive lurking intention from a role perspective. By integrating these two aspects, we can delve into the impacts of role stresses engendered by the distinctive social environment on SNS, thus expanding our comprehension of platform-based features within the social context of SNS. Through this role perspective, we also arrive at conclusions that challenge established beliefs, offering fresh insights into the role of platform-based features within the SNS social environment.

Second, we have successfully identified boundary spanning as a second-order construct comprising two distinct dimensions: *parallelism* and *reprocessability*. Through a comparative analysis between offline social environments and SNS social environments, we have posited that SNS liberate users from the constraints of time and space, thereby fostering frequent boundary-spanning social activities and their consequent challenges. Drawing from an exhaustive literature review, we have identified two pertinent media capabilities, namely, parallelism and reprocessability, to encapsulate this unique social context inherent to SNS.

Parallelism empowers users to engage in social interactions unhindered by geographical limitations, facilitating social engagement irrespective of physical location. On the other hand, reprocessability empowers users to partake in social interactions at their convenience, obviating temporal limitations. The convergence of these two media capabilities amalgamates a user's various social circles and blurs the demarcations between distinct social roles. Boundary spanning was traditionally measured by refining an instrument developed by [Aiken and Hage \(1972\)](#) and [Miles \(1974\)](#). The final instrument was measured with the following dimensions: (1) internal boundary spanning: internal verbal and written communication; (2) external boundary spanning: external verbal and written communication, e.g. customers, suppliers, trade, consultants; (3) formal boundary spanning: formal meetings and formal written communication occurring, e.g. reports or data from others; (4) informal boundary spanning: informal face-to-face or telephone conversations or informal written communication, e.g. informal notes, memos and the like; (5) selected aspects: includes travel and coordinating activities. This measurement is not suitable for the context of SNS because it is very different from traditional marketing organizations. It would be useful to understand how instrumental boundary spanning operates within the context of SNS. As our literature review shows, existing research on SNS lurking behaviors has overlooked the role of media contextual factors. We have identified parallelism and reprocessability as two primary constructs that underpin boundary spanning from a media capabilities perspective. This insight introduces a novel measurement for distinguishing and quantifying the unique social environment of SNS, thereby providing fresh insights into the body of SNS literature. In particular, it highlights the influence of SNS contextual-specific cues on user behaviors.

Third, anchored in the burnout theoretical model and integrated with a role perspective, this study stands as one of the pioneering efforts in elucidating the mechanism behind the maladaptive lurking intention observed among SNS users. We enhance the current research conceptualization on the SNS lurking phenomenon by emphasizing that not all lurking behaviors are active or justified, particularly when examining role dynamics. Our empirical tests further contribute to the existing body of research on the SNS lurking mechanism by highlighting the pivotal influence of context-specific cues (such as social network size and boundary spanning), maladaptive cognition (such as role conflict and role overload) and distorted affect (i.e. SNS fatigue). These insights significantly enrich the SNS lurking literature, facilitating a deeper comprehension of the decision-making processes behind users' lurking intentions. The integration of role theory and the burnout theory also theoretically extends these two theories.

6.3 Managerial implications

The results of this study provide compelling evidence that both the size of a user's social network and the act of boundary spanning on SNS can lead to perceived role conflict and role overload. As users form more social connections, their networks become more diverse. Coupled with the boundary-spanning characteristic of SNS, this can result in inevitable role conflict and role overload. Consequently, users may find it increasingly challenging and taxing to navigate their social interactions effectively, which can lead to SNS fatigue and an inclination to lurk rather than actively participate. The sustainability of SNS platforms heavily relies on users actively posting and engaging. Thus, these findings have significant practical implications for SNS providers. Providers should exercise caution when recommending new contacts to users or sharing potential contact information. While these recommendations aim to enhance user interactions, merely increasing the number of connections can sometimes have adverse effects, especially in terms of role stresses. Furthermore, SNS providers should consider offering tools that are context-aware and can

assist users in managing their expanding and complex social networks. Although current SNS, such as WM and Facebook, enable users to categorize their contacts into distinct groups to mitigate potential role stresses, yet this function frequently fails. The requirement for users to select continuously and manually who can view their posts can be an exhaustive process, particularly as the size and complexity of their contact lists increase. Thus, more effective and user-friendly mechanisms, like AI-based applications, should be developed to learn users' current SNS posting behavior and suggest more customized controls over users' role stresses, while recommending their postings to a particular group that they intend to see. Additionally, it's essential for SNS platforms to manage and, where necessary, limit technological attributes such as parallelism and reprocessability, which might promote maladaptive behaviors. The ultimate goal should be to foster a healthy and sustainable user experience on these platforms.

6.4 Limitations and future research directions

While this research offers valuable theoretical and practical insights, our study comes with several limitations, pointing to intriguing avenues for future research. First, our data were predominantly sourced from WMs users in China. To gain a broader understanding, subsequent research could extend data collection to users of various SNS like Facebook or Twitter in different countries. Second, our cross-sectional approach restricts our ability to delve deeply into actual lurking behaviors. A future field experiment observing users' longitudinal social interactions on SNS could provide insights into more objective behavioral and procedural outcomes. Third, our research focused on the unique SNS social environment, primarily considering the impact of users' social network size and the boundary-spanning technical feature. However, further studies are needed to delve into other specific contextual cues of SNS. For instance, understanding the actual number of social roles a user assumes on SNS, and determining the importance of different roles within various social network contexts would be beneficial. This is especially relevant as the stresses associated with different roles, shaped by varying organizational norms and cultures, can differ significantly. Lastly, the impact of individual user differences was not fully considered in this study. Such as users with large social networks who have few frequent interactions may differ significantly from those with smaller networks but more frequent interactions. Future research could benefit from exploring how different user characteristics influence their lurking intention.

7. Conclusions

SNS empower users with the robust ability to connect with others, significantly enhancing their management of diverse social connections and online personas. However, as users' social network sizes expand, coupled with the inherent boundary-spanning feature of SNS, they encounter unprecedented role stresses. These stresses can compromise efficient relationship management and even tarnish their online image. Observing this paradox, we delved into the influence of context-specific cues in the SNS social environment on users' lurking intentions. Anchored in the burnout theoretical model and integrated a role perspective, this study stands as one of the pioneering efforts in elucidating the mechanism behind the maladaptive lurking intention observed among SNS users. Our findings indicate that both an increase in social network size and boundary spanning significantly amplify role conflict and role overload. These, in turn, intensify SNS fatigue, which escalates users' intentions to lurk. This study offers a fresh viewpoint for comprehending lurking behaviors on SNS, furnishing practical insights for platform providers. Additionally, it paves the way for future research into the deeper mechanisms driving SNS lurking phenomenon.

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Context	Topology	Theories	Main findings	Sources
Online communities	Technical features supporting communities are the main motivations for lurking	Commitment theory, organization support theory, IT-enabled public goods theory, Construal level theory	Reciprocity, the richness of contents, perceived moderator enthusiasm, speed of response, extroversion, trust, socialization, type of community, technical design flaws, collective benefits, familiarity with the community, group identity, and privacy and safety of the online group are identified as indicators of lurking	Zhu and Dawson (2024) , Badreddine and Blount (2021) , Yang et al. (2017) , Amichai-Hamburger et al. (2016) , Phang et al. (2015) , Rafaeli et al. (2004) , Chen (2004)
	Psychological factors are the main motivations for lurking	Self-determination theory, Uses and gratifications theory	Psychological distance, cognitive distance, emotional distance, cultural distance, brand loyalty and gratification are identified as indicators of lurking	Bozkurt et al. (2020) , Fernandes and Castro (2020) , Amichai-Hamburger et al. (2016) , Mo and Coulson (2010) , Nonnecke and Preece (2001)
	Personalities are the main motivations for lurking	Self-determination theory, Curiosity-drive theory, Social learning theory	Personality dispositions, tendency toward social loafing, time available and self-efficacy, epistemic curiosity, free-riding, legitimate peripheral participation, microlearning, knowledge sharing barriers, loneliness, optimism, depression, health-related quality of life, anonymously, not needing to post, cultural capital, and self-esteem are identified as indicators of lurking	Fernandes and Castro (2020) , Schneider et al. (2013) , Neelen and Fetter (2011) , Mo and Coulson (2010) , Panciera et al. (2010) , Preece et al. (2008) , Soroka and Rafaeli (2006) , Nonnecke et al. (2004) , Chen (2004)
SNS	Psychological factors are the main motivations for lurking	Conservation of resources theory, Motivation theory, Social penetration theory, Interpersonal relationship perspective	Emotional exhaustion, social media ostracism, social interaction anxiety, disappointment, social interaction, entertainment, computer anxiety, and intimacy are identified as indicators of lurking	Ali et al. (2024) , Hong et al. (2023) , Liu et al. (2020) , De Veirman et al. (2016) , Osatuyi (2015) , Rau et al. (2008)
	Personalities are the main motivations for lurking	Listening theory, Transaction model of stress, Self-discrepancy theory, Privacy calculus theory, Personal trait theory, Engagement theory	Leisure, current affairs, curiosity, anxiety, opinion detachment, cognitive apprenticeship, political participation, neglected unintended audience concern, privacy risk belief, conscientiousness, Machiavellianism, empty nest feelings, poor online communication skills, lack of confidence, lack of a sense of belonging, personal trait of overall engagement, and disposition toward a particular topic are identified as indicators of lurking	Adjin-Tettey and Garman (2023) , Zhang et al. (2021) , Ortiz et al. (2018) , Seigfried-Spellar and Lankford (2018) , Tanis et al. (2017) , Shafie et al. (2016) , Muller (2012)
Source(s): Authors' own creation				

Appendix 2

Constructs	Items	References
Parallelism	[Par1] WeChat Moment allows me to communicate with multiple parties concurrently [Par2] WeChat Moment allows simultaneous sending of messages to multiple recipients [Par3] WeChat Moment allows simultaneous receipt of messages from multiple senders [Par4] WeChat Moment allows concurrent conversations to occur	Adapted from He and Yang, (2016)
Reprocessability	[Rep1] WeChat Moment allows me to reexamine and reprocess previously sent content during the interaction [Rep2] WeChat Moment allows me to reexamine and reprocess previously sent content after the interaction [Rep3] WeChat Moment allows me to reexamine and consider previously sent content for developing understanding and additional consideration [Rep4] WeChat Moment provides a memory that can remind me on early discussion contents	Adapted from He and Yang (2016)
Role Conflict	[RC1] I frequently receive incompatible expectations from two or more parties on WeChat Moment [RC2] I do things on WeChat Moment that are apt to be accepted by one person and not accepted by others [RC3] I often have to try to balance two or more conflicting activities on WeChat Moment [RC4] I sometimes have to break a rule or norm to complete the thing I would like to do on WeChat Moment	Adapted from Rutner et al. (2008) ; Zhang et al. (2011)
Role Overload	[RO1] I have too many roles to play on WeChat Moment [RO2] I feel overburdened by different roles I have taken on WeChat Moment [RO3] I have been given too much role expectation from other peers on WeChat Moment [RO4] Different expectations from others on WeChat Moment make me too tired or irritable to participate in or enjoy the activities on WeChat Moment [RO5] The amounts of roles I have to play interfere with the actual things I would like to do on WeChat Moment [RO6] I feel I have too many roles to comfortably handle on WeChat Moment	Adapted from Zhang et al. (2011) ; Ho et al. (2003) ; Peterson (1995)
SNS Fatigue	[SNSF1] Sometimes I feel drained from using WeChat Moment [SNSF1] Sometimes I feel worn out from using WeChat Moment [SNSF1] Sometimes I feel tired when using WeChat Moment [SNSF1] Sometimes I feel bored when using WeChat Moment	Adapted from Zhang et al. (2016)

Table A2.
The survey instrument

(continued)

Constructs	Items	References
Lurking Intention	[LI1] In the future, I will decrease the number of my posts on WeChat Moment [LI2] In the future, I will decrease the frequency of my posts on WeChat Moment [LI3] In the future, I will post on WeChat Moment far less than today [LI4] If I could, I would not post on WeChat Moment anymore	Adapted from Edelmann, (2013) ; Zhang et al. (2016)
Privacy Concern	[PC1] I am concerned that the information I submit on WeChat Moment could be misused [PC2] I am concerned that a person can find private information about me on WeChat Moment [PC3] I am concerned about submitting information on WeChat Moment, because of what others might do with it [PC4] I am concerned about submitting information on WeChat Moment, because it could be used in a way I did not foresee	Adapted from Dinev and Hart (2006)
Fashion Consciousness	[FC1] When I must choose between the two, I usually dress for fashion, not for comfort [FC2] An important part of my life and activities is dressing smartly [FC3] A person should try to dress in style	Adapted from Malhotra et al. (2006)
Source(s): Authors' own creation		

Table A2.

Appendix 3

Items	VIF
Par1	3.018
Par2	2.485
Par3	3.365
Par4	2.583
ReP1	2.929
ReP2	3.100
ReP3	2.108
ReP4	1.910
RC1	1.756
RC2	1.760
RC3	1.909
RC4	1.882
RO1	1.516
RO2	2.223
RO3	2.039
RO4	1.643
RO5	2.846
RO6	2.431
SNSF1	1.692
SNSF2	3.287
SNSF3	3.249
SNSF4	2.486
LI1	3.971
LI2	4.389
LI3	3.973
LI4	1.418
PC1	1.790
PC2	2.428
PC3	3.018
PC4	2.883

Table A3.
Variance inflation
factors (VIFs)

Note(s): Parallelism (Par); Reprocessability (Rep); Role conflict (RC); Role overload (RO); SNS fatigue (SNSF);
Lurking intention (LI); Privacy concern (PC)
Source(s): Authors' own creation

Appendix 4

Relationships	Social network size as a dichotomous variable (low:0 vs high:1)	Difference (Frequency_high - Frequency_low)
Social Network Size → Role Conflict	0.238**	0.144 ^{ns}
Social Network Size → Role Overload	0.25**	0.205*
Boundary Spanning → Role Conflict	0.116*	-0.187 ^{ns}
Boundary Spanning → Role Overload	0.133**	-0.157 ^{ns}
Role Conflict → SNS Fatigue	0.229***	-0.059 ^{ns}
Role Conflict → Lurking Intention	0.1*	0.217*
Role Overload → SNS Fatigue	0.389***	0.029 ^{ns}
Role Overload → Lurking Intention	0.111*	-0.233*
SNS Fatigue → Lurking Intention	0.407***	0.08 ^{ns}
Gender → Lurking Intention	0.054 ^{ns}	-0.156 ^{ns}
Age → Lurking Intention	-0.068 ^{ns}	0.082 ^{ns}
Education → Lurking Intention	0.042 ^{ns}	0.066 ^{ns}
Career → Lurking Intention	-0.032 ^{ns}	0.005 ^{ns}
Income → Lurking Intention	0.004 ^{ns}	-0.05 ^{ns}
Tenure → Lurking Intention	-0.015 ^{ns}	0.032 ^{ns}
Frequency → Lurking Intention	0.079*	0.093 ^{ns}
Privacy Concern → Lurking Intention	0.17***	0.001 ^{ns}

Note(s): *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns, non-significant
Source(s): Authors' own creation

Table A4.
Robustness tests of the results

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